# Word Embeddings

*Presented By*

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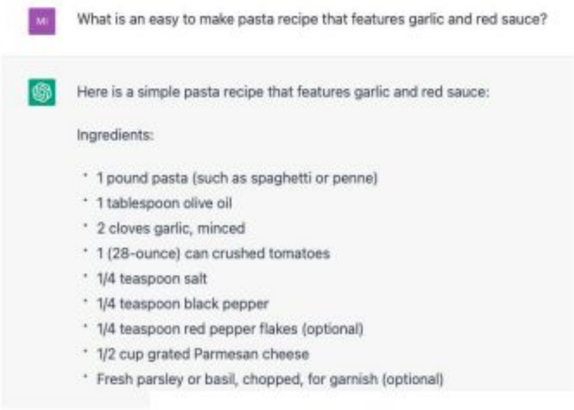
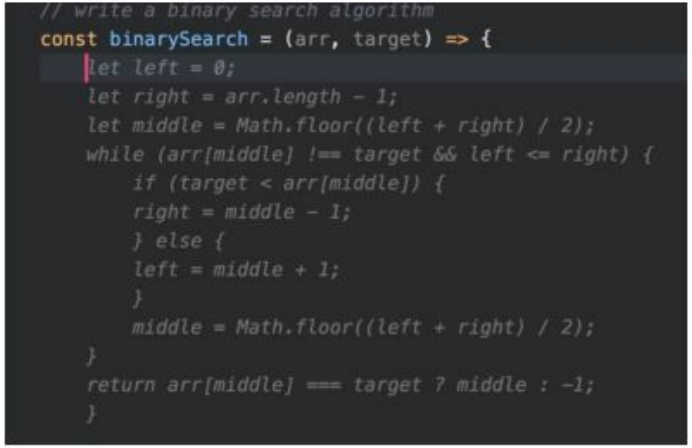




# Representing words by their context

**Distributional hypothesis**: words that occur in similar contexts tend to have similar meanings

J.R.Firth 1957

- "You shall know a word by the company it keeps"
- One of the most successful ideas of modern statistical NLP!

| | | |
|---|---|---|
| *...government debt problems turning into* | ***banking*** | *crises as happened in 2009...* |
| *...saying that Europe needs unified* | ***banking*** | *regulation to replace the hodgepodge...* |
| *...India has just given its* | ***banking*** | *system a shot in the arm...* |

These context words will represent *banking*.

# Distributional hypothesis

“tejuino”

C1: A bottle of ___ is on the table.

C2: Everybody likes ___.

C3: Don’t have ___ before you drive.

C4: We make ___ out of corn.

# Distributional hypothesis

C1: A bottle of ___ is on the table.

C2: Everybody likes ___.

C3: Don't have ___ before you drive.

C4: We make ___ out of corn.

| | C1 | C2 | C3 | C4 |
|---|---|---|---|---|
| tejuino | 1 | 1 | 1 | 1 |
| loud | 0 | 0 | 0 | 0 |
| motor-oil | 1 | 0 | 0 | 0 |
| tortillas | 0 | 1 | 0 | 1 |
| choices | 0 | 1 | 0 | 0 |
| wine | 1 | 1 | 1 | 0 |

"words that occur in similar contexts tend to have similar meanings"

# How to represent words?

N-gram language models

$$P(w \mid \text{it is 76 F and})$$

It is 76 F and ____.

[0.0001, 0.1, 0, 0, 0.002, ..., 0.3, ..., 0]
red sunny

Text classification

$$P(y = 1 \mid x) = \sigma(\theta^{\intercal} w + b)$$

I like this movie. 👍

$w^{(1)}$ [0, 1, 0, 0, 0, ..., 1, ..., 1]

I don't like this movie. 👎

$w^{(2)}$ [0, 1, 0, 1, 0, ..., 1, ..., 1]
don't

# Representing words as discrete symbols

In traditional NLP, we regard words as discrete symbols:

hotel, conference, motel — a localist representation

one 1, the rest 0's

Words can be represented by one-hot vectors:

hotel = [0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0]

motel = [0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0]

Vector dimension = number of words in vocabulary (e.g., 500,000)

Challenge: How to compute similarity of two words?

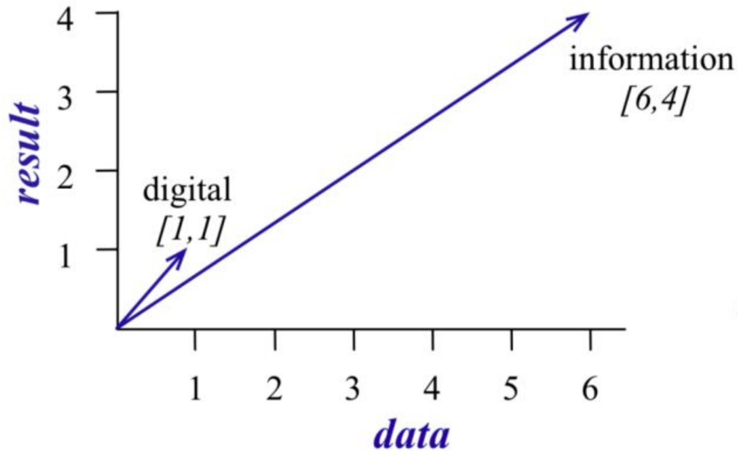
# Words as vectors

- We'll build a new model of meaning focusing on similarity
  - Each word is a vector
  - Similar words are "nearby in space"

- A first solution: we can just use context vectors to represent the meaning of words!
  - word-word co-occurrence matrix:

| | aardvark | computer | data | pinch | result | sugar | ... |
|---|---|---|---|---|---|---|---|
| apricot | 0 | 0 | 0 | 1 | 0 | 1 | |
| pineapple | 0 | 0 | 0 | 1 | 0 | 1 | |
| digital | 0 | 2 | 1 | 0 | 1 | 0 | |
| information | 0 | 1 | 6 | 0 | 4 | 0 | |

# Words as vectors




$$cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$cos(\mathbf{u}, \mathbf{v}) = \frac{\sum_{i=1}^{V} u_i v_i}{\sqrt{\sum_{i=1}^{V} u_i^2} \sqrt{\sum_{i=1}^{V} v_i^2}}$$

What is the range of cos( $\cdot$ )?

# Words as vectors

Problem: not all counts are equal, words can randomly co-occur

- Solution: re-weight by how likely it is for the two words to co-occur by simple chance
- PPMI = Positive Pointwise Mutual Information

$$\text{PPMI}(w,c) = \max(\log_2 \frac{P(w,c)}{P(w)P(c)}, 0)$$

| | computer | data | result | pie | sugar |
|---|---|---|---|---|---|
| **cherry** | 2 | 8 | 9 | 442 | 25 |
| **strawberry** | 0 | 0 | 1 | 60 | 19 |
| **digital** | 1670 | 1683 | 85 | 5 | 4 |
| **information** | 3325 | 3982 | 378 | 5 | 13 |

| | computer | data | result | pie | sugar |
|---|---|---|---|---|---|
| **cherry** | 0 | 0 | 0 | 4.38 | 3.30 |
| **strawberry** | 0 | 0 | 0 | 4.10 | 5.51 |
| **digital** | 0.18 | 0.01 | 0 | 0 | 0 |
| **information** | 0.02 | 0.09 | 0.28 | 0 | 0 |

# Sparse vs dense vectors

- Still, the vectors we get from word-word occurrence matrix are sparse (most are 0's) & long (vocabulary size)

- Alternative: we want to represent words as **short** (50-300 dimensional) & **dense** (real-valued) vectors
    - The focus of this lecture
    - The basis of all the modern NLP systems

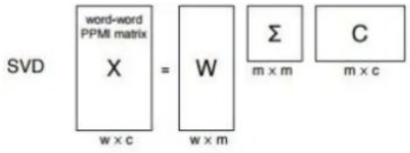
# Dense vectors

$$\text{employees} = \begin{pmatrix} 0.286 \\ 0.792 \\ -0.177 \\ -0.107 \\ 10.109 \\ -0.542 \\ 0.349 \\ 0.271 \\ 0.487 \end{pmatrix}$$

# Why dense vectors?

- Short vectors are easier to use as features in ML systems
- Dense vectors may generalize better than storing explicit counts
- They do better at capturing synonymy
  - $w_1$ co-occurs with "car", $w_2$ co-occurs with "automobile"

- Different methods for getting dense vectors:
  - Singular value decomposition (SVD)
  - word2vec and friends: "learn" the vectors!



# Word2vec

- Input: a large text corpora, *V*, *d*
  - V: a pre-defined vocabulary
  - d: dimension of word vectors (e.g. 300)
  - Text corpora:
    - Wikipedia + Gigaword 5: 6B
    - Twitter: 27B
    - Common Crawl: 840B
- Output: $f : V \to \mathbb{R}^d$

$$v_{\text{cat}} = \begin{pmatrix} -0.224 \\ 0.130 \\ -0.290 \\ 0.276 \end{pmatrix} \quad v_{\text{dog}} = \begin{pmatrix} -0.124 \\ 0.430 \\ -0.200 \\ 0.329 \end{pmatrix}$$

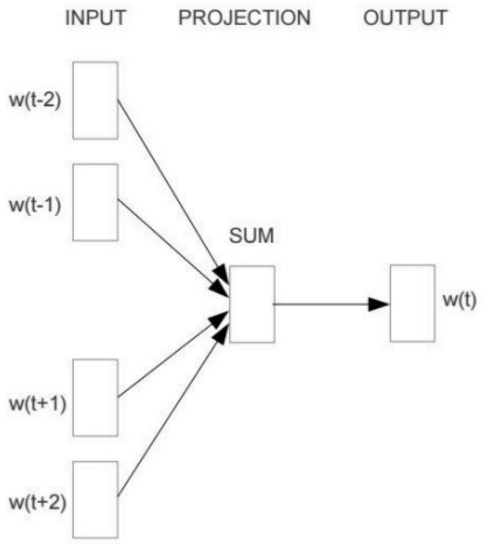
$$v_{\text{the}} = \begin{pmatrix} 0.234 \\ 0.266 \\ 0.239 \\ -0.199 \end{pmatrix} \quad v_{\text{language}} = \begin{pmatrix} 0.290 \\ -0.441 \\ 0.762 \\ 0.982 \end{pmatrix}$$

# Word2vec

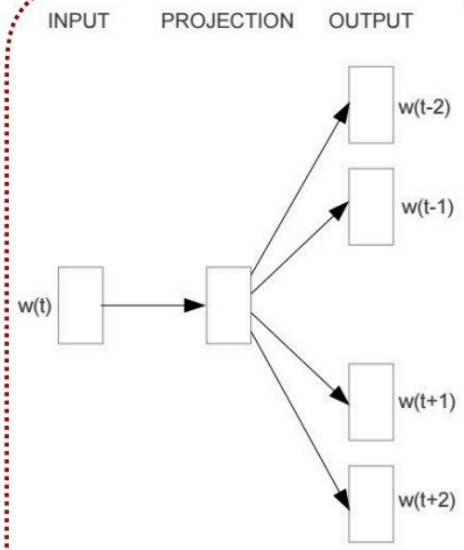
word = “sweden”

| Word | Cosine distance |
|---|---|
| norway | 0.760124 |
| denmark | 0.715460 |
| finland | 0.620022 |
| switzerland | 0.588132 |
| belgium | 0.585835 |
| netherlands | 0.574631 |
| iceland | 0.562368 |
| estonia | 0.547621 |
| slovenia | 0.531408 |

# Word2vec



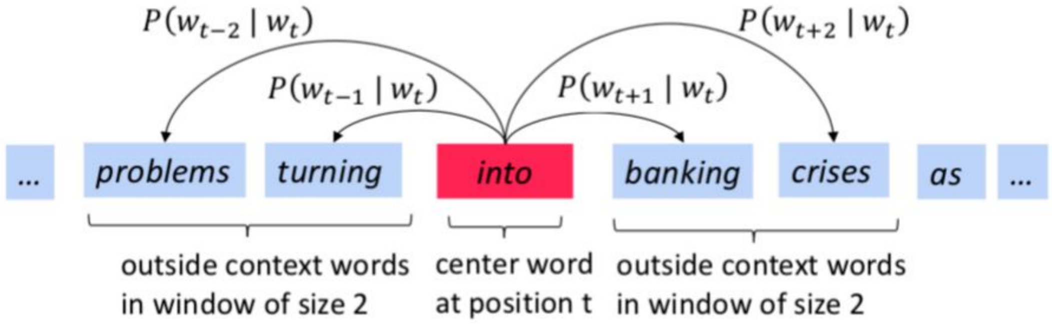

Continuous Bag of Words (CBOW)



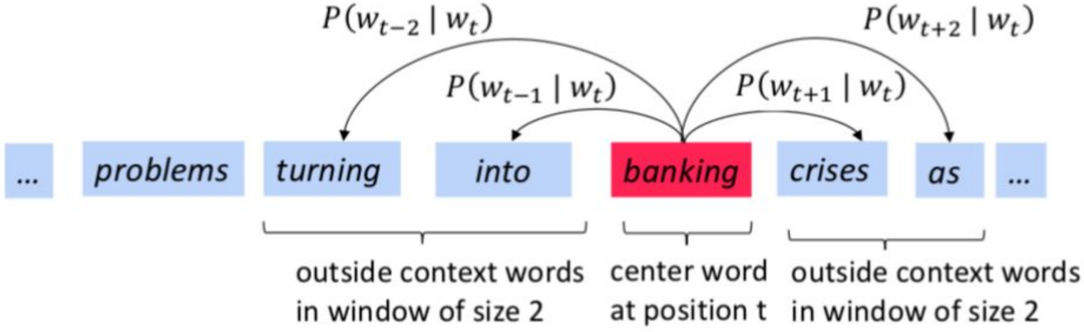

Skip-grams

# Skip-gram

- The idea: we want to use words to **predict** their context words
- Context: a fixed window of size $2m$



# Skip-gram


$P(w_{t-2} \mid w_t)$
$P(w_{t-1} \mid w_t)$
$P(w_{t+1} \mid w_t)$
$P(w_{t+2} \mid w_t)$
...
problems
turning
into
banking
crises
as
...
outside context words in window of size 2
center word at position t
outside context words in window of size 2

# Skip-gram: objective function

- For each position $t = 1,2,\ldots T$, predict context words within context size m, given center word $w_j$:

all the parameters to be optimized

$$\mathcal{L}(\theta) = \prod_{t=1}^{T} \prod_{-m \leq j \leq m, j \neq 0} P(w_{t+j} \mid w_t; \theta)$$

- The objective function $J(\theta)$ is the (average) negative log likelihood:

$$J(\theta) = -\frac{1}{T} \log \mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^{T} \sum_{-m \leq j \leq m, j \neq 0} \log P(w_{t+j} \mid w_t; \theta)$$

# How to define $P(w_{t+j} \mid w_t; \theta)$?

- We have two sets of vectors for each word in the vocabulary

$$\mathbf{u}_i \in \mathbb{R}^d \text{ : embedding for target word } i$$

$$\mathbf{v}_{i'} \in \mathbb{R}^d \text{ : embedding for context word } i'$$

- Use inner product $\mathbf{u}_i \cdot \mathbf{v}_{i'}$ to measure how likely word *i* appears with context word *i'*, the larger the better

"softmax" we learned last time!

$$P(w_{t+j} \mid w_t) = \frac{\exp(\mathbf{u}_{w_t} \cdot \mathbf{v}_{w_{t+j}})}{\sum_{k \in V} \exp(\mathbf{u}_{w_t} \cdot \mathbf{v}_k)}$$

$\theta = \{\{\mathbf{u}_k\}, \{\mathbf{v}_k\}\}$ are all the parameters in this model!

# How to train the model

Calculating all the gradients together!

$$\theta = \{\{\mathbf{u}_k\}, \{\mathbf{v}_k\}\}$$

$$J(\theta) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{-m\leq j\leq m, j\neq 0} \log P(w_{t+j} \mid w_t; \theta) \qquad \nabla_\theta J(\theta) = ?$$

Q: How many parameters are in total?

We can apply stochastic gradient descent (SGD)!

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta J(\theta)$$

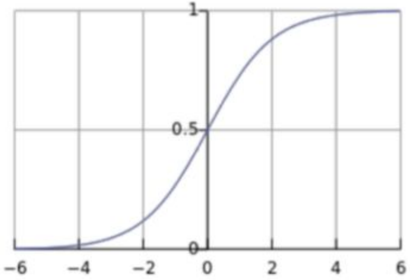
# Skip-gram with negative sampling (SGNS)

**Idea:** recast problem as binary classification!

- Target word is positive example
- All words not in context are negative

$$P(D = 1 \mid t, c) = \sigma(\mathbf{u}_t \cdot \mathbf{v}_c)$$

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$




**positive examples +**

| t | c |
|---|---|
| apricot | tablespoon |
| apricot | of |
| apricot | jam |
| apricot | a |

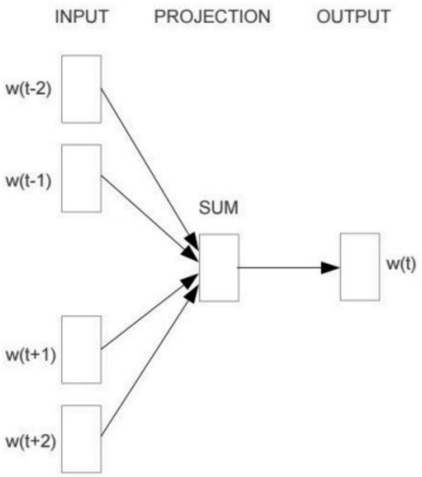
**negative examples -**

| t | c | t | c |
|---|---|---|---|
| apricot | aardvark | apricot | seven |
| apricot | my | apricot | forever |
| apricot | where | apricot | dear |
| apricot | coaxial | apricot | if |

To compute loss, pick K random words as negative examples:

$$J(\theta) = -P(D = 1 \mid t, c) - \frac{1}{K} \sum_{i=1}^{K} P(D = 0 \mid t_i, c)$$

# Continuous Bag of Words (CBOW)




$$L(\theta) = \prod_{t=1}^{T} P\left(w_t \mid \{w_{t+j}\}, -m \leq j \leq m, j \neq 0\right)$$

$$\bar{\mathbf{v}}_t = \frac{1}{2m} \sum_{-m \leq j \leq m, j \neq 0} \mathbf{v}_{t+j}$$

$$P(w_t \mid \{w_{t+j}\}) = \frac{\exp(\mathbf{u}_{w_t} \cdot \bar{\mathbf{v}}_t)}{\sum_{k \in V} \exp(\mathbf{u}_k \cdot \bar{\mathbf{v}}_t)}$$

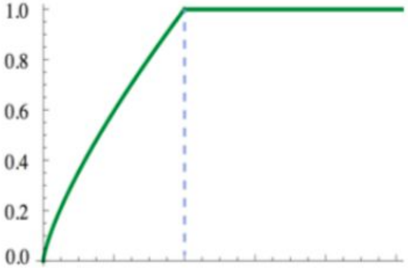
# GloVe: **Glo**bal **Ve**ctors

- Let's take the global co-occurrence statistics: $X_{i,j}$

$$J = \sum_{i,j=1}^{V} f\left(X_{ij}\right)\left(w_i^T \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij}\right)^2$$

- Training faster
- Scalable to very large corpora

$f \sim$




(Pennington et al, 2014): GloVe: Global Vectors for Word Representation

# GloVe: **Glo**bal **Ve**ctors

Nearest words to
frog:

1. frogs
2. toad
3. litoria
4. leptodactylidae
5. rana
6. lizard
7. eleutherodactylus

litoria

leptodactylidae

rana

eleutherodactylus

(Pennington et al, 2014): GloVe: Global Vectors for Word Representation

# FastText: Sub-Word Embeddings

- Similar as Skip-gram, but break words into n-grams with n = 3 to 6

where:

3-grams: <wh, whe, her, ere, re>

4-grams: <whe, wher, here, ere>

5-grams: <wher, where, here>

6-grams: <where, where>

- Replace $\mathbf{u}_i \cdot \mathbf{v}_j$ by $\sum_{g \in n\text{-grams}(w_i)} \mathbf{u}_g \cdot \mathbf{v}_j$

- More to come! Contextualized word embeddings

(Bojanowski et al, 2017): Enriching Word Vectors with Subword Information

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# Trained word embeddings available

- word2vec: https://code.google.com/archive/p/word2vec/
- GloVe: https://nlp.stanford.edu/projects/glove/
- FastText: https://fasttext.cc/

**Download pre-trained word vectors**

- Pre-trained word vectors. This data is made available under the Public Domain Dedication and License v1.0 whose full text can be found at: http://www.opendatacommons.org/licenses/pddl/1.0/.
  - Wikipedia 2014 + Gigaword 5 (6B tokens, 400K vocab, uncased, 50d, 100d, 200d, & 300d vectors, 822 MB download): glove.6B.zip
  - Common Crawl (42B tokens, 1.9M vocab, uncased, 300d vectors, 1.75 GB download): glove.42B.300d.zip
  - Common Crawl (840B tokens, 2.2M vocab, cased, 300d vectors, 2.03 GB download): glove.840B.300d.zip
  - Twitter (2B tweets, 27B tokens, 1.2M vocab, uncased, 25d, 50d, 100d, & 200d vectors, 1.42 GB download): glove.twitter.27B.zip
- Ruby script for preprocessing Twitter data

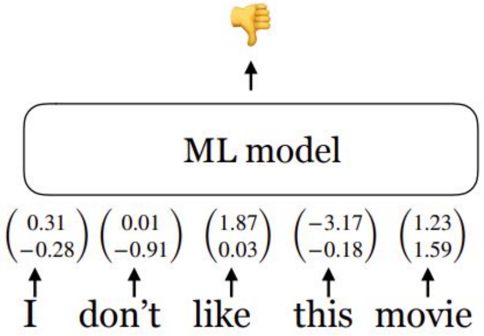
Differ in algorithms, text corpora, dimensions, cased/uncased...

# Evaluating Word Embeddings

# Extrinsic vs intrinsic evaluation

Extrinsic evaluation

- Let's plug these word embeddings into a real NLP system and see whether this improves performance
- Could take a long time but still the most important evaluation metric




Intrinsic evaluation

- Evaluate on a specific/intermediate subtask
- Fast to compute
- Not clear if it really helps the downstream task

# Intrinsic evaluation

**Word similarity**

Example dataset: wordsim-353
353 pairs of words with human judgement
http://www.cs.technion.ac.il/~gabr/resources/data/wordsim353/

| Word 1 | Word 2 | Human (mean) |
|---|---|---|
| tiger | cat | 7.35 |
| tiger | tiger | 10 |
| book | paper | 7.46 |
| computer | internet | 7.58 |
| plane | car | 5.77 |
| professor | doctor | 6.62 |
| stock | phone | 1.62 |
| stock | CD | 1.31 |
| stock | jaguar | 0.92 |

Cosine similarity:

$$\cos(\boldsymbol{u}_i, \boldsymbol{u}_j) = \frac{\boldsymbol{u}_i \cdot \boldsymbol{u}_j}{||\boldsymbol{u}_i||_2 \times ||\boldsymbol{u}_j||_2}.$$

Metric: Spearman rank correlation

# Intrinsic evaluation

**Word Similarity**

| Model | Size | WS353 | MC | RG | SCWS | RW |
|---|---|---|---|---|---|---|
| SVD | 6B | 35.3 | 35.1 | 42.5 | 38.3 | 25.6 |
| SVD-S | 6B | 56.5 | 71.5 | 71.0 | 53.6 | 34.7 |
| SVD-L | 6B | 65.7 | <u>72.7</u> | 75.1 | 56.5 | 37.0 |
| CBOW† | 6B | 57.2 | 65.6 | 68.2 | 57.0 | 32.5 |
| SG† | 6B | 62.8 | 65.2 | 69.7 | <u>58.1</u> | 37.2 |
| GloVe | 6B | <u>65.8</u> | <u>72.7</u> | <u>77.8</u> | 53.9 | <u>38.1</u> |
| SVD-L | 42B | 74.0 | 76.4 | 74.1 | 58.3 | 39.9 |
| GloVe | 42B | <u>**75.9**</u> | <u>**83.6**</u> | <u>**82.9**</u> | <u>**59.6**</u> | <u>**47.8**</u> |
| CBOW* | 100B | 68.4 | 79.6 | 75.4 | 59.4 | 45.5 |

# Intrinsic evaluation

**Word analogy**

man: woman ≈ king: ?

$$\arg\max_i \left(\cos(\mathbf{u}_i, \mathbf{u}_b - \mathbf{u}_a + \mathbf{u}_c)\right)$$

semantic

Chicago:Illinois≈Philadelphia: ?

syntactic

bad:worst ≈ cool: ?

More examples at

http://download.tensorflow.org/data/questions-words.txt

# What can go wrong with word embeddings?

- What’s wrong with learning a word’s “meaning” from its usage?
- What data are we learning from?
- What are we going to learn from this data?

# What do we mean by bias?

- Identify *she* - *he* axis in word vector space, project words onto this axis

| | Extreme *she* occupations | |
|---|---|---|
| 1. homemaker | 2. nurse | 3. receptionist |
| 4. librarian | 5. socialite | 6. hairdresser |
| 7. nanny | 8. bookkeeper | 9. stylist |
| 10. housekeeper | 11. interior designer | 12. guidance counselor |

| | Extreme *he* occupations | |
|---|---|---|
| 1. maestro | 2. skipper | 3. protege |
| 4. philosopher | 5. captain | 6. architect |
| 7. financier | 8. warrior | 9. broadcaster |
| 10. magician | 11. figher pilot | 12. boss |

Bolukbasi et al. (2016)

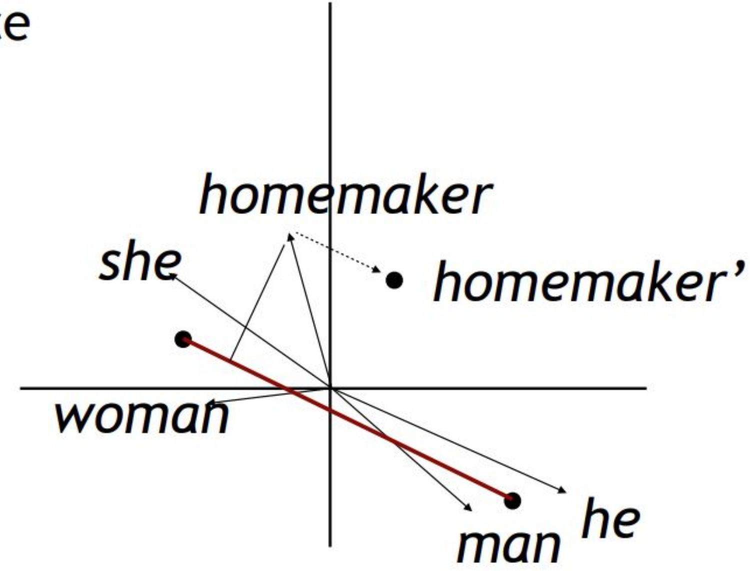
- Nearest neighbor of (b - a + c)

| **Racial Analogies** | |
|---|---|
| black → homeless | caucasian → servicemen |
| caucasian → hillbilly | asian → suburban |
| asian → laborer | black → landowner |
| **Religious Analogies** | |
| jew → greedy | muslim → powerless |
| christian → familial | muslim → warzone |
| muslim → uneducated | christian → intellectually |

Manzini et al. (2019)

# Debiasing

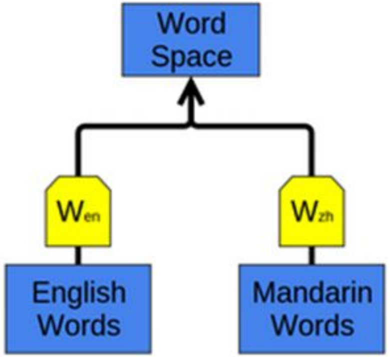
- Identify gender subspace with gendered words
- Project words onto this subspace
- Subtract those projections from the original word




Bolukbasi et al. (2016)

# Word embedding applications

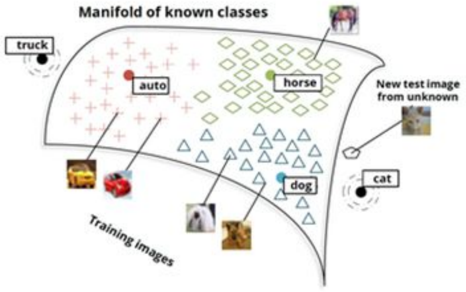
- Can learn to map multiple kinds of data into a single representation.
- E.g., bilingual English and Mandarin Chinese word-embedding as in Socher *et al.* (2013a).
- Embed as above, but words that are known as close translations should be close together.
- Words we *didn't know* were translations end up close together!
- Structures of two languages get pulled into alignment.



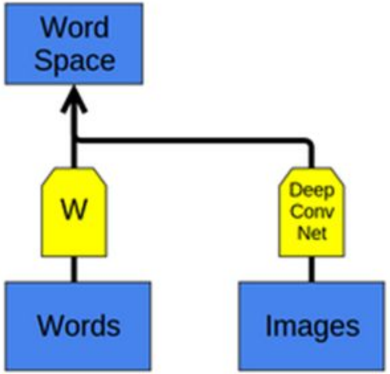

http://colah.github.io/posts/2014-07-NLP-RNNs-Representations/

# Word embedding applications

- Can apply to get a joint embedding of words and images or other multi-modal data sets.
- New classes map near similar existing classes: e.g., if ‘cat’ is unknown, cat images map near dog.




(Socher *et al.* (2013b))

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# Language Modeling: predict the next word

**Assign probabilities to text.**

Given a sequence $(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)$, we want to **maximize** $P(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)$.

$$P(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T) = P(\mathbf{x}_1)P(\mathbf{x}_2|\mathbf{x}_1)P(\mathbf{x}_3|\mathbf{x}_1, \mathbf{x}_2)P(\mathbf{x}_4|\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)\ldots P(\mathbf{x}_T|\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_{T-1})$$

P(I like cats because they look cute) = P(I) P(like | I) P(cats | I like) P(as | I like cats) P(they | I like cats because)

P(look | I like cats because they) P(cute | I like cats because they look)

Predict the next word given current text!

# *n*-Gram Language Model

*n*-Gram: chunk of *n* consecutive words

**Uni-gram:** "I" "like" "cats" "as" "they" "look" "cute"

**Bi-gram:** "I like" "like cats" "cats as" "as they" …

**Tri-gram:** "I like cats" "like cats as" "cats as they" …

**Count** the frequency of each *n*-grams and predict next word!

**Assume** each word only depends on previous *n - 1* words.

$$P(\mathbf{x}_t|\mathbf{x}_1,\ldots,\mathbf{x}_{t-1}) = P(\mathbf{x}_t|\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1})$$
$$= \frac{\text{count}(\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1},\mathbf{x}_t)}{\text{count}(\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1})}$$

In *bi-gram* LM

P(I like cats as they look cute) = P(I) P(like | I) P(cats | like) P(as | cats) P(they | because) P(look | they) P(cute | look)

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# *n*-Gram Language Model

*n*-Gram: chunk of *n* consecutive words

**Uni-gram:** "I" "like" "cats" "as" "they" "look" "cute"

**Bi-gram:** "I like" "like cats" "cats as" "as they" …

**Count** the frequency of each *n*-grams and predict next word!

**Tri-gram:** "I like cats" "like cats as" "cats as they" …

**Assume** each word only depends on previous *n - 1* words.

$$P(\mathbf{x}_t|\mathbf{x}_1,\ldots,\mathbf{x}_{t-1}) = P(\mathbf{x}_t|\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1}) = \frac{\operatorname{count}(\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1},\mathbf{x}_t)}{\operatorname{count}(\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1})}$$

In *bi-gram* LM

P(I like cats as they look cute) = P(I) P(like | I) P(cats | like) P(as | cats) P(they | because) P(look | they) P(cute | look)

# *n*-Gram Language Model: issue

*n*-Gram: chunk of *n* consecutive words

**Uni-gram:** "I" "like" "cats" "as" "they" "look" "cute"

**Bi-gram:** "I like" "like cats" "cats as" "as they" …

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$$P(\mathbf{x}_t|\mathbf{x}_1,\ldots,\mathbf{x}_{t-1}) = P(\mathbf{x}_t|\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1})$$
$$= \frac{\mathrm{count}(\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1},\mathbf{x}_t)}{\mathrm{count}(\mathbf{x}_{t-n+1},\ldots,\mathbf{x}_{t-1})}$$

**Increasing *n* provides contextual information, but exponentially increases the size of the counting table!**

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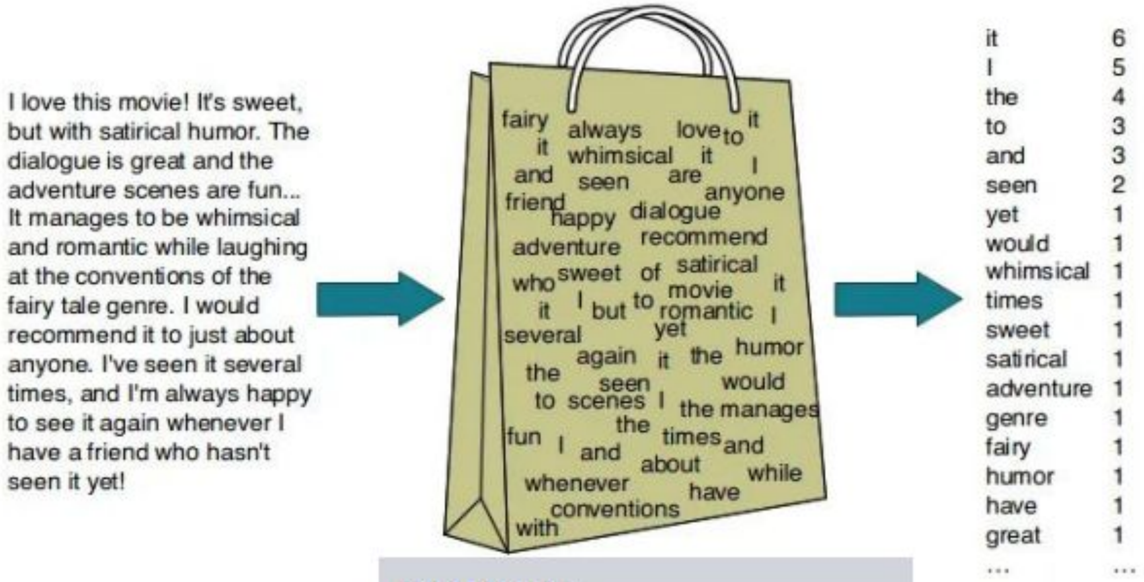
# Bag of Words (to represent documents)




Drawbacks:

- High dimensionality
- No semantic information

https://koushik1102.medium.com/nlp-bag-of-words-and-tf-idf-explained-fd1f49dce7c4

# Document similarity?

<u>document 1</u>

**Obama**
**speaks**
to
the
**media**
in
**Illinois**

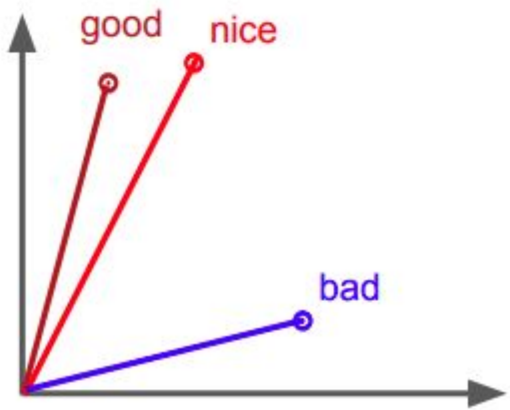
Documents have no words in common.
How can we quantify that they convey similar meanings?
(Assume B. Obama is president.)

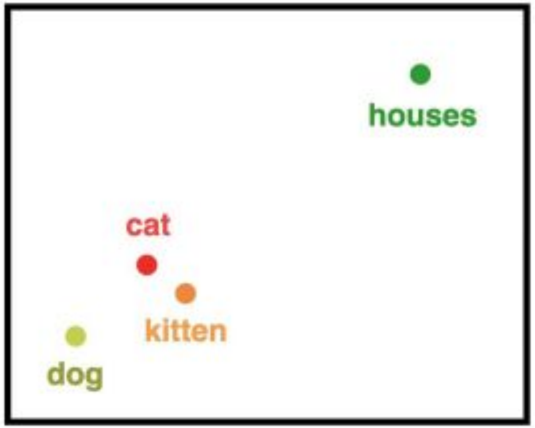
<u>document 2</u>

The
**President**
**greets**
the
**press**
in
**Chicago**

# Semantic similarity

- Motivation
  - Put words into vectors so we can measure the similarity between words
  - Use cosine similarity



# Why Do We Need Word Embeddings?

- Why Do We Need Word Embeddings?
  - Numerical Input
  - Shows Similarity and Distance


| | living being | feline | human | gender | royalty | verb | plural |
|---|---|---|---|---|---|---|---|
| cat | 0.6 | 0.9 | 0.1 | 0.4 | -0.7 | -0.3 | -0.2 |
| kitten | 0.5 | 0.8 | -0.1 | 0.2 | -0.6 | -0.5 | -0.1 |
| dog | 0.7 | -0.1 | 0.4 | 0.3 | -0.4 | -0.1 | -0.3 |
| houses | -0.8 | -0.4 | -0.5 | 0.1 | -0.9 | 0.3 | 0.8 |
| man | 0.6 | -0.2 | 0.8 | 0.9 | -0.1 | -0.9 | -0.7 |
| woman | 0.7 | 0.3 | 0.9 | -0.7 | 0.1 | -0.5 | -0.4 |
| king | 0.5 | -0.4 | 0.7 | 0.8 | 0.9 | -0.7 | -0.6 |
| queen | 0.8 | -0.1 | 0.8 | -0.9 | 0.8 | -0.5 | -0.9 |

**embedding using features of words**

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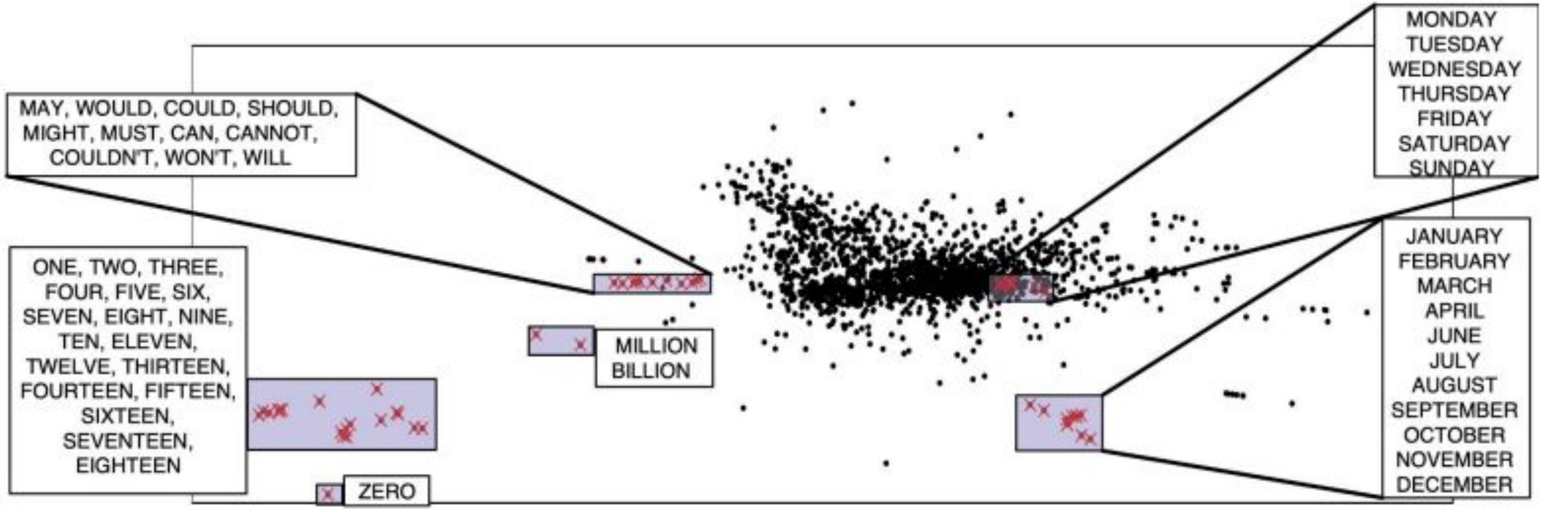
# What are word embeddings

- What are Word Embeddings?
    - vector representations of words that capture semantic relationships
    - Latent Semantic Analysis / Indexing [S. Deerwester et al 1988]




[Blitzer et al. Neurips, 2004]

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# Efficient Estimation of Word Representations in Vector Space

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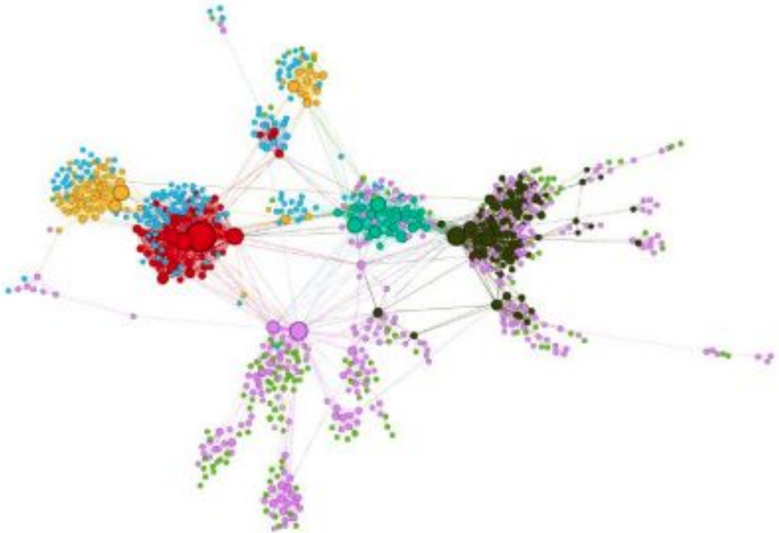
## Abstract

We propose two novel model architectures for computing continuous vector representations of words from very large data sets. The quality of these representations is measured in a word similarity task, and the results are compared to the previously best performing techniques based on different types of neural networks. We observe large improvements in accuracy at much lower computational cost, i.e. it takes less than a day to learn high quality word vectors from a 1.6 billion words data set. Furthermore, we show that these vectors provide state-of-the-art performance on our test set for measuring syntactic and semantic word similarities.

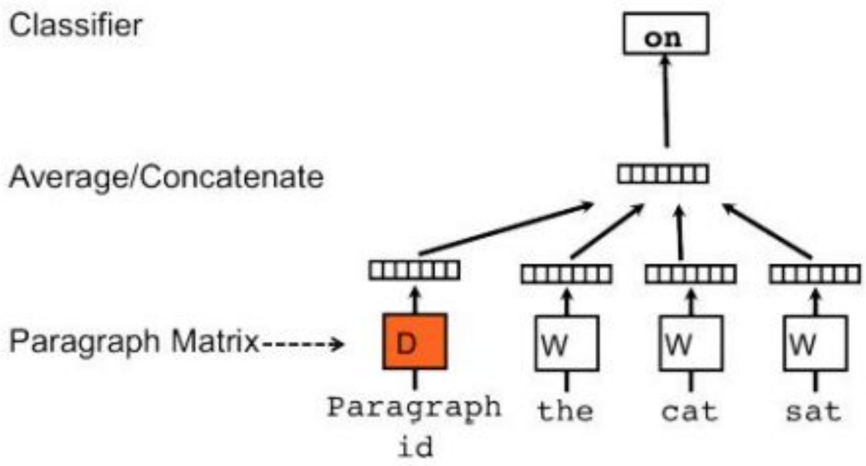
# X 2 vec

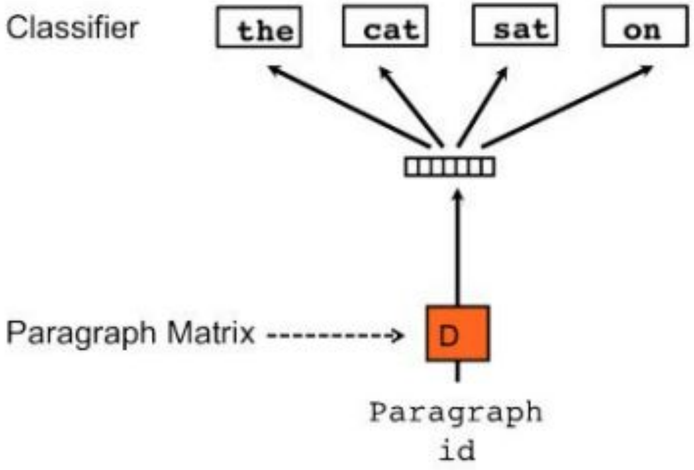
- Generate vector representations (embeddings) for various data types
- Examples:
  - Word2Vec
  - Doc2Vec
  - Node2Vec
  - Item2Vec
  - Sent2Vec



# Doc2Vec

- A vector to represent a paragraph, regardless of length
  - embeddings for paragraph and words
  - Applications: Document classification, sentiment analysis, recommendation systems, and information retrieval

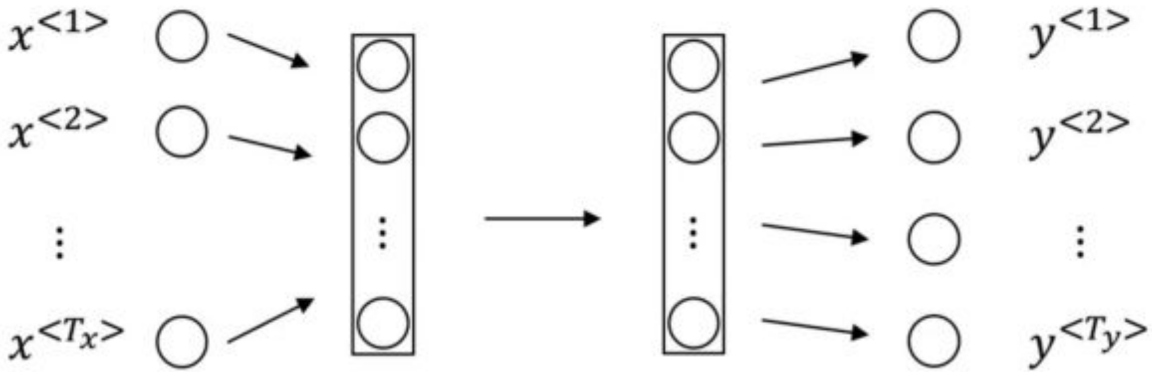


# Problems with word2vec

- Words with multiple meanings only have one representation
  - eg. **bank** of river or **bank** of money
  - Need contextual information
- Limited Context
  - only trained on words within the context window

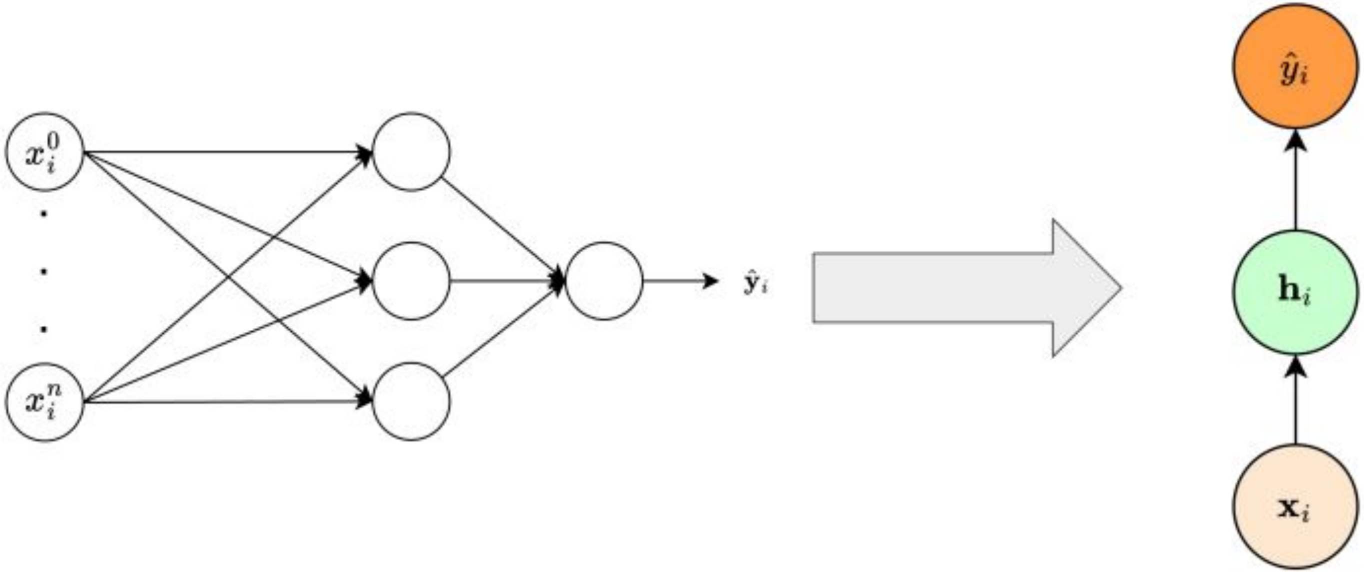
# How to use word vectors with neural networks?



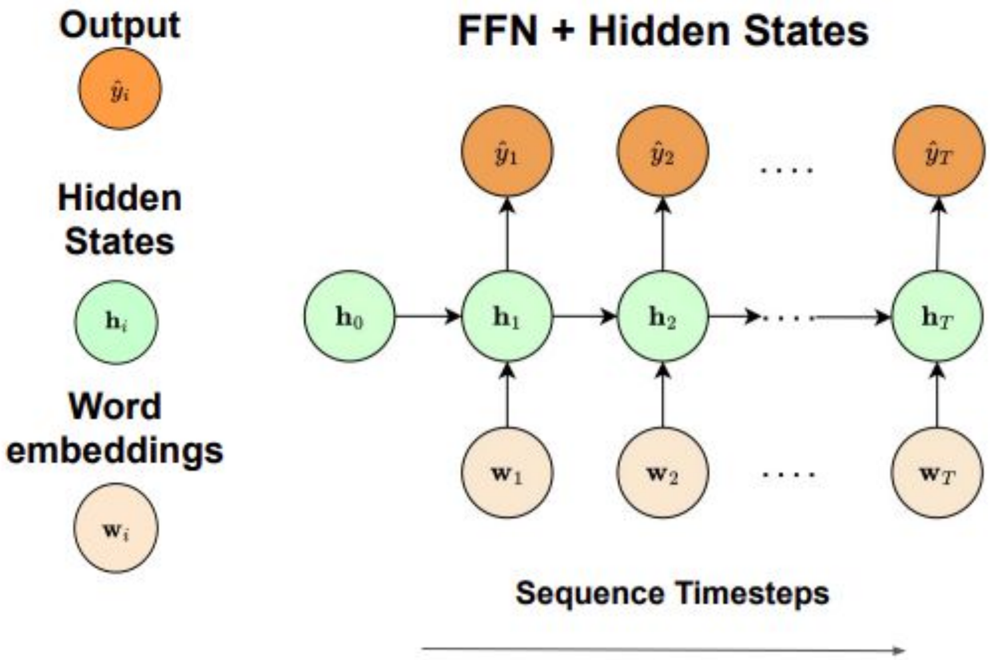

- Inputs and outputs don’t have fixed lengths
- Features are not shared

# Let's simplify!

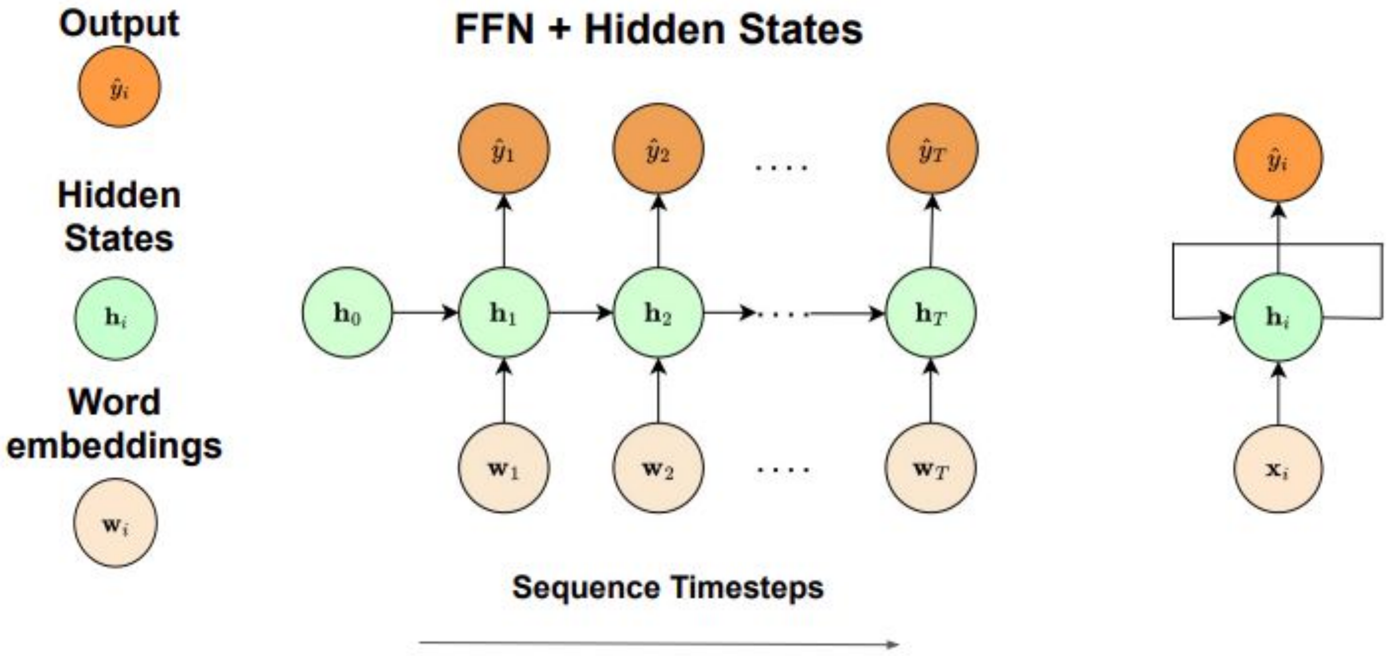
What if we have a single word and a single output?



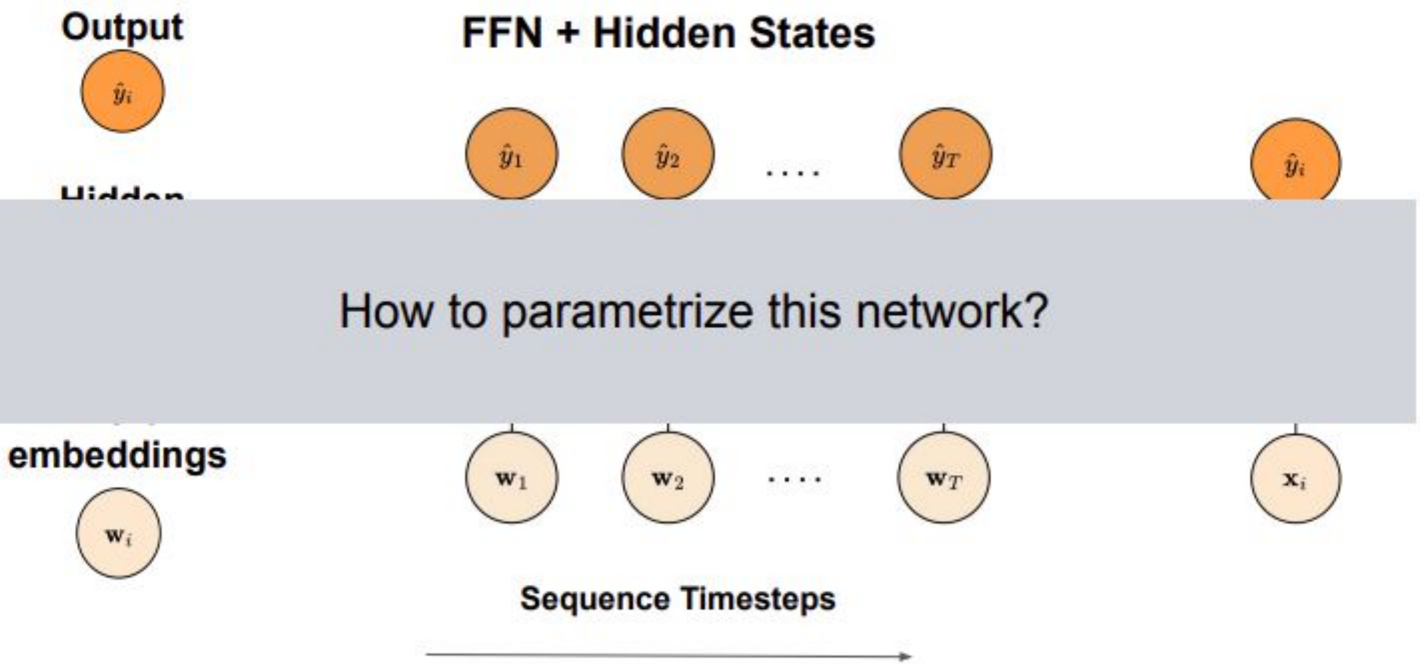
# Recurrent neural network (RNN)



# Recurrent neural network (RNN)



# Recurrent neural network (RNN)




How to parametrize this network?

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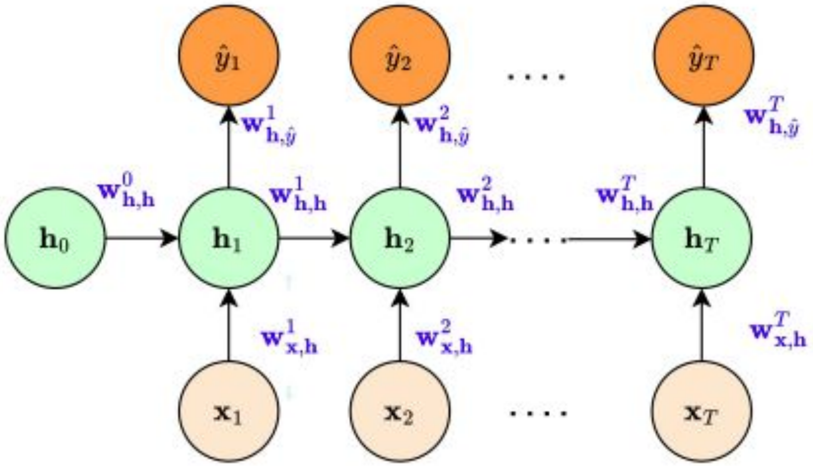
# Parameterize RNN




- Too many parameters if we have a long sequence!
- Longer sequence parameters will not receive many updates
- What if sequence lengths vary?

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# RNN w/ parameter-sharing

Simple fix: use **the same parameters** across different timesteps.